Name Surname: .2022

ID:

Signature:

CSE4065 Fall 2022 Quiz III (Duration: 50 min)

Q1) (20 pts.) Consider the following alignment:

```
AGCT - TCA
GG - - G TTA
```

a) What is the score of this alignment if the **match score** is **2**, the **mismatch penalty** is **1**, and the indel penalty is affine with a **gap opening penalty** of **2** and a **gap extension penalty** of **1**?

$2.3 - 2 - 2.2 - 1 = \boxed{-1}$ ✓

b) Considering the same penalties, can you find a better alignment using the same patterns? Calculate the score of your alignment.

```
AGCTTCA
GGGTT-A
```

match: 4
mismatch: 2
open: 1
ext: 0

$4.2 - 2 - 2.1 \boxed{= 4}$ ✓

c) Apply local alignment, what is the score? Also show the alignment.

```
A - [GCTTCA]
G - [GGTT-A]
```

score: $4.2 - 1 - 2 = 5$
8 - 3 ✓

Q2) (20 pts.) What are the differences between sequence alignment and genome rearrangement?

Assume you are given two sequences. When do you apply sequence alignment and genome rearrangement?

Q3) (20 pts.) Compute the number of steps required by the given algorithm to sort the following permutation (i.e., to transform it into the identity permutation.) Show your work in each step.

5 3 4 2 6 1

a) Greedy Algorithm – Simple Reversal Sort

initial: 5 3 4 2 6 1

1) 1 6 2 4 3 5

2) 1 2 6 4 3 5

3) 1 2 3 4 6 5

4) 1 2 3 4 5 6 ✓

b) Improved Break Point Reversal Sort

initial: 0 | 5 | 3 4 | 2 | 6 | 1 | 7

1) 0 1 | 6 | 2 | 4 3 | 5 | 7

2) 0 1 | 6 | 2 3 4 5 | 7

3) 0 1 | 6 5 4 3 2 | 7

4) 0 1 2 3 4 5 6 7

Q4) (20 pts.) You are given two chromosomes and asked whether you can obtain the second chromosome from the first chromosome. The sequences of the synteny blocks that are available in the chromosomes are given below. You can use any algorithm. Show all your work at each step.

+5 −3 −2 −1 −4

Chromosome I = GGATTCGCGAAAGTCCCGTATATGCAGCA

Chromosome II= CGTATACCTGAAAGCGCTTACGACGTGGA

Synteny Blocks: −1 +2 +3 +4 +5

1 = ATATGC ✓

2 = CCTGA ✓

3 = AAGCGCTT ✓

4 = ACGACGT ✓

5 = GGA ✓

Initial: +5 −3 −2 −1 −4

1) +1 +2 +3 −5 −4

2) +1 +2 +3 +4 +5

3) −1 +2 +3 +4 +5

Q5) (20 pts.) Given the following permutations P and Q:

P = (+a +b +c +d +e +f)

Q= (-b +d -f -e +a -c)

Show the breakpoint graph Breakpoint(P, Q) and Calculate the 2-break distance between P and Q, d(P,Q). Show the steps to transform P to Q using 2-breaks.

